

# **Continuous Age-Dependent Analysis of Somatic Mutation Profiles and Survival in PDAC**

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## **Abstract**

### **Background**

Traditional oncology research frequently utilizes static bins to categorize patient age, which leads to risk of masking continuous molecular shifts across a lifespan. In Pancreatic Ductal Adenocarcinoma (PDAC), understanding how somatic driver mutations and survival metrics evolve continuously with age is vital for advancing research in the field and improving treatment options.

### **Methods**

Clinical and genomic profiles from a population of 670 PDAC patients, including 514 death events, were evaluated. Age at diagnosis was used as an unmanipulated continuous covariate. Non-linear proportion tracking evaluated KRAS driver variant frequencies across continuous age decade midpoints. Overall survival was modeled using continuous Cox Proportional Hazards (CoxPH) regression alongside categorical Kaplan-Meier curves for comparison.

### **Results**

KRAS driver variants showed a non-linear selection pattern over the lifespan. The KRAS G12D variant declined from 60% prevalence at age 35 to 40% at age 75, whereas KRAS G12V increased from 20% to nearly 50% by age 95. Binned Kaplan-Meier quartiles produced heavily overlapping survival curves, while the continuous Cox modeling showed distinct, sequential risk gradients across the lifespan. In terms of structural genomic aberrations, Fraction Genome Altered with a coefficient of 0.89 and a p-value of 0.01, and Loss of Heterozygosity with a coefficient of -0.81 and a p of less than 0.005, proved to be the primary drivers of mortality risk.

### **Conclusion**

When treating age as an unmanipulated continuous variable, age-dependent mutational selection and smooth survival gradients that were obscured by traditional binning are revealed. This highlights the need of continuous modeling frameworks to be implemented in the field of cancer genomics.

## Background

PDAC is one of the most lethal malignancies around the world. It is currently a large problem in the medical field as a result of its late diagnosis and limited responsiveness to standard chemotherapeutic treatments. PDAC represents 90% of the total pancreatic malignancies and has an overall five-year survival rate of below 10% (1). The reason for this is because of the delay in diagnosis and often misdiagnosis early in the stages. In one study, 98 of 313 patients had an initial misdiagnosis, the most common being: gallbladder disease, gastroesophageal reflux disease, and peptic ulcer disease (2).

PDAC can occur in patients of all demographics and age groups. However, it is often cited that this cancer is found rarely in patients below 50 years of age and there is limited information regarding risk factors (3). In oncology patient age is frequently treated like this, on a fixed threshold such as 50 years of age or “early-onset” and “late-onset”. However, aging is a continuous process and mutations are not likely to follow such a rigid system. By relying on this, researchers risk missing linear and non-linear patterns that may occur within patients incrementally over their lifespan. This project will look to address how somatic mutation frequencies and alteration types in the TP53 and KRAS genes evolve across a lifespan.

The specific problem area that will be investigated by this study is the simplification of clinical patient age during the profiling of tumor driver mutation in Pancreatic Ductal Adenocarcinoma (PDAC) patients. Age is one of the most prevalent risk factors in oncology (4). Specifically, this study focuses on *KRAS* and *TP53*, as *KRAS* acts as the primary initiating oncogenic driver present in over 90% of PDAC tumors, while *TP53* serves as a critical tumor suppressor driving late-stage progression and genomic instability. However, it is often categorized into bins, early-onset and late onset (young and old), which can lead to a serious loss of statistical data. Understanding a more precise relationship between age and tumor genetics is crucial for the field. It provides a biological basis for why different age groups may experience distinct rates of disease progression, which is necessary for developing personalized clinical interventions.

The primary purpose of this study is to implement advanced data science modeling strategies to analyze the relationship between patient age and PDAC driver mutations. By using age as a continuous variable, this project will map the precise progression of alterations across a lifespan to address one of the main limitations found in oncology literature. It is hypothesized that the frequency of primary TP53 and KRAS driver mutations will scale non-linearly with continuous age, displaying distinct changes during specific life decades as opposed to uniform distribution.

Ultimately, challenging the current approach to cancer research can help to highlight the risk in the current medical landscape. The results of this project can inform more flexible and patient appropriate screening guidelines. This will overall lead to more personalized treatment plans and better informed decisions to increase health outcomes.

## Methods

This study plans to utilize two primary data types that will be extracted from public archives: somatic genomic sequencing data and clinical profile data. This data type combination is rationalized by the study's goal to align tumor mechanisms against continuous clinical patient outcomes. This project will be entirely quantitative in nature. This is because the research questions depend upon evaluating statistical significance, creating regression models, and calculating continuous hazard metrics across a large population dataset.

The primary data will come from The Cancer Genome Atlas (TCGA) and the cBioPortal for Cancer Genomics, accessing the Pancreatic Adenocarcinoma (PAAD) datasets. TCGA is chosen because it represents the gold standard for open access genomic and clinical cancer data, while cBioPortal provides the technical blueprint that will be required to filter and isolate multidimensional variables.

The data will be collected and cleaned utilizing the resources provided by the cBioPortal and programming software. Firstly, Comprehensive clinical and mutation data will be downloaded from cBioPortal from the TCGA PAAD section. Then, the data will be cleaned by identifying and removing any entries that lack an age at diagnosis or overall survival metrics.

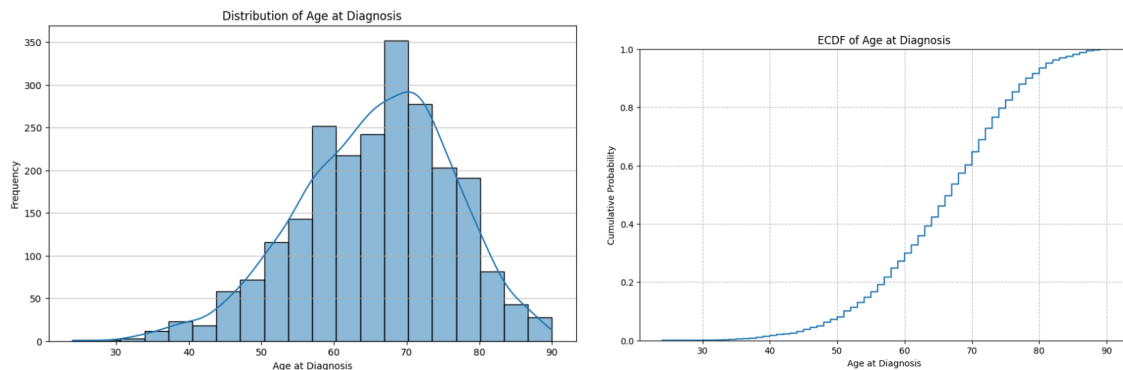
The target study population includes all available data on patients diagnosed with Pancreatic Ductal Adenocarcinoma. The cBioPortal dataset includes a total of 2336 unique PDAC patient records which will be the sample size. Demographically, the sample spans a continuous age range from as young as 24 to 90 years of age, covering diverse biological backgrounds, survival rates, and ancestry. The data for TCGA is collected through a rigorous, multi-step process. First, hospitals and tissue banks collect surgically removed tumor tissue and matched normal control samples from patients. Then, the samples are sent to a central Biospecimen Core Resource (BCR) where pathologists review them. Next, the BCR extracts the DNA and RNA and analytes are sent to Genome Sequencing Centers (GSC) and Genome Characterization Centers (GCC), where they generate the various types of data (5). Overall, because this project relies on this open-access data from TCGA via the cBioPortal, there are no ethical risks.

In terms of measurement, the study will use a combination of multivariate regression, continuous smoothing models, and proportional hazards modeling. These methods can bypass the need for static categorical bins, allowing age to remain an unmanipulated, independent continuous variable. This will be necessary to reveal any biological trends that may be present in patients of varying ages with PDAC.

The main data science methodology and tool that will be utilized in this study is python (Pandas and Statsmodels) for exploratory data visualization. Multivariate regression is very important because of its ability to estimate the coefficients of the linear equation, involving one or more independent variables that best predict the value of the dependent variable which should be quantitative (6). Additionally, continuous smoothing models are often used in medical studies to model survival rates or to detect spatial

variations in disease spread (7). In this study in particular it will be very useful in modeling survival rates of PDAC patients of varying age demographics. Finally proportional hazard models are vital in estimating the ratio of the "hazard" (the risk of an event) between different groups while adjusting for multiple risk factors simultaneously (8). This is useful in this study because we can compare the most dangerous risk factors when it comes to PDAC, whether it be age, smoking history, or race. These methods were selected over traditional descriptive tools, like chi-squared and basic T-tests, because they can better capture non-linear, gradual transitions over time.

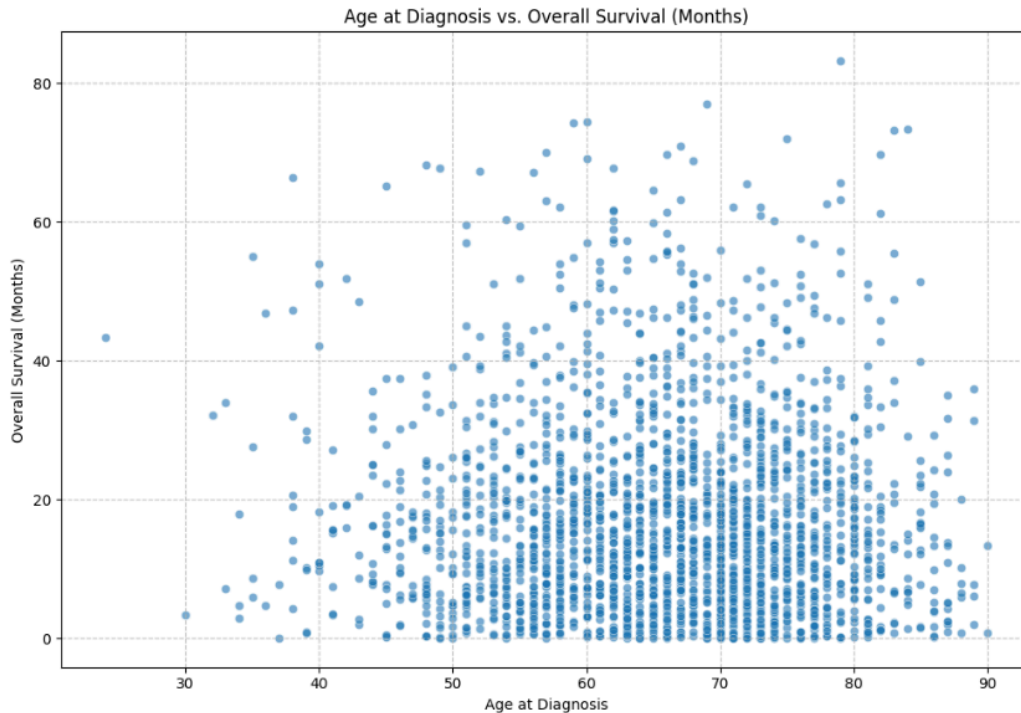
Exploratory data analysis was performed as a baseline for the project.



**Figure 1. Age at Diagnosis Distribution across the PDAC Study Population.** Left: Histogram and kernel density estimation (KDE) plot showing the continuous distribution of age at diagnosis across the study cohort (N = 670). Right: Empirical Cumulative Distribution Function (ECDF) illustrating the continuous progression of patient diagnoses across a 66-year observational window.

The population consists of N = 2,336 unique pancreatic ductal adenocarcinoma patient records. Records were systematically filtered based on complete clinical and genomic profile availability. Cases were excluded if they lacked confirmed age at diagnosis, overall survival metrics, complete somatic mutation calls for target driver genes (*KRAS* and *TP53*), or structural copy-number/MSI metadata. Following quality control filtering, the final population consisted of N = 670 unique patient records with 514 recorded death events. Analysis of the "Age at Diagnosis" feature confirms a continuous distribution of patients spanning from the lowest age of 24, to a maximum of 90 years old. This is a 66 year observational window that provides a necessary span across adult life decades to map any incremental alterations that may be seen. This will ultimately allow for better analysis in different age groups, and more specialized treatments.

To visually check that clinical and biological outcomes do not break cleanly at certain age points like 50, a bivariate scatter plot was created mapping age at diagnosis and the overall survival rate in months.



**Figure 2. Bivariate Relationship Between Continuous Age at Diagnosis and Overall Survival.** Scatter plot displaying individual patient overall survival in months against age at diagnosis. Reveals no clustering or relevant patterns around the traditional 50-year cutoff which highlights the continuous, non-linear distribution of clinical outcomes across patient lifespan.

The distribution reveals a dense, continuous cluster of data points concentrated between the ages of 45 and 85. This is reflective of the expected peak incidence of late-stage pancreatic cancers. Tracking the data across the traditional 50-year cutoff shown in most literature reveals much more uniformity. Patient survival limits span from 0 to over 80 months on both sides of the mark. No sudden visual discontinuity or alterations are observed at or around the age of 50.

The lack of an abrupt threshold shows the necessity of the use of localized scatter plot smoothing and continuous hazard modeling. Forcing this group of points into categorical bins destroys the variance present across life's decades, confirming that a continuous profile is required to capture the true, non-linear survival patterns.

Overall, the project objectives are clearly defined, and entirely achievable. The required clinical variables of age, smoking history, survival, and the genomic parameters (TP53 and KRAS structures) are fully populated within the accessible TCGA dataset. The statistical tools chosen are well-suited to handle this specific research, and will go a long way in advancing the research in the realm of cancer genomics.

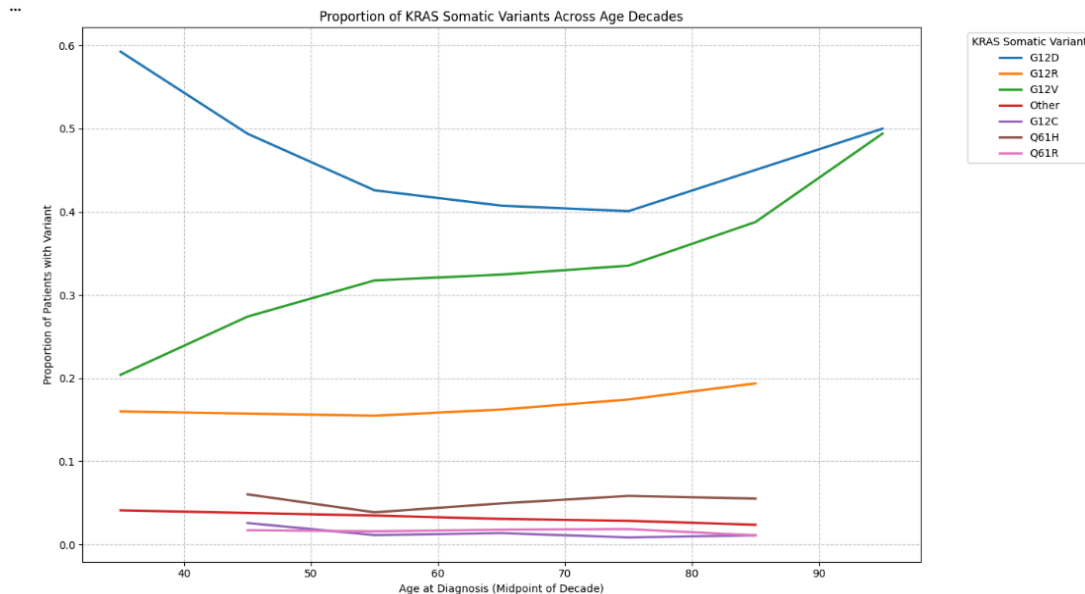
## Results

### Cohort Characteristics and Baseline Age Distribution

After filtering the original study population, the final count was  $N = 670$  unique patient records containing 514 recorded deaths. The patient age features a span of a continuous 66-year observational window ranging from 24 to 90 years old. Bivariate scatter plotting of continuous age against overall survival in months showed a dense cluster of data points concentrated between the ages of 45 and 85 years old, with patient survival spanning from 0 to 80 months on both sides of the 50-year mark.

### Lifespan Mapping of Somatic Mutation Profiles

To investigate how somatic mutation frequencies and individual alteration types evolve across the adult lifespan, individual *KRAS* driver variants were mapped against the continuous age midpoints of each patient decade (Figure 3).



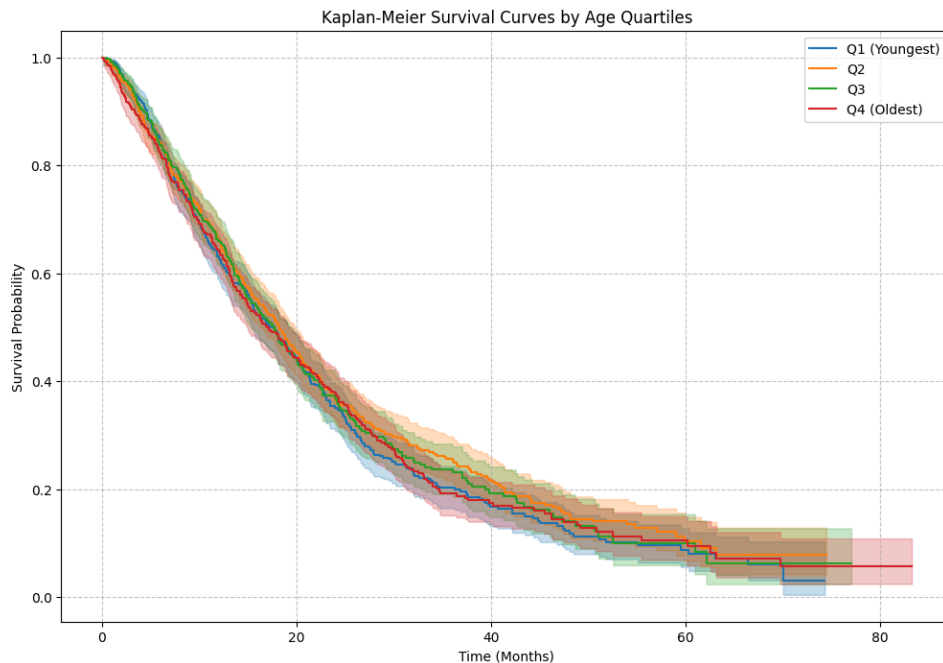
**Figure 3. Proportion of *KRAS* Somatic Variants Across Lifespan Decades.**

Lifespan trajectory mapping of specific *KRAS* driver variants plotted against continuous age midpoints per decade. Lines indicate non-linear selection shifts, highlighting diverging trends between G12D (blue) and G12V (green) across adulthood.

The G12D variant occurred in approximately 60% of patients at the age 35 midpoint, decreased steadily to approximately 40% by age 75, and showed a minor upward inflection in the following decades. Additionally, the G12V variant increased across the lifespan, rising from approximately 20% prevalence at age 35 to approximately 50% by age 95. Other localized variants, including G12R, remained uniform across the observational window, maintaining a baseline between 15% and 20% across all age decades.

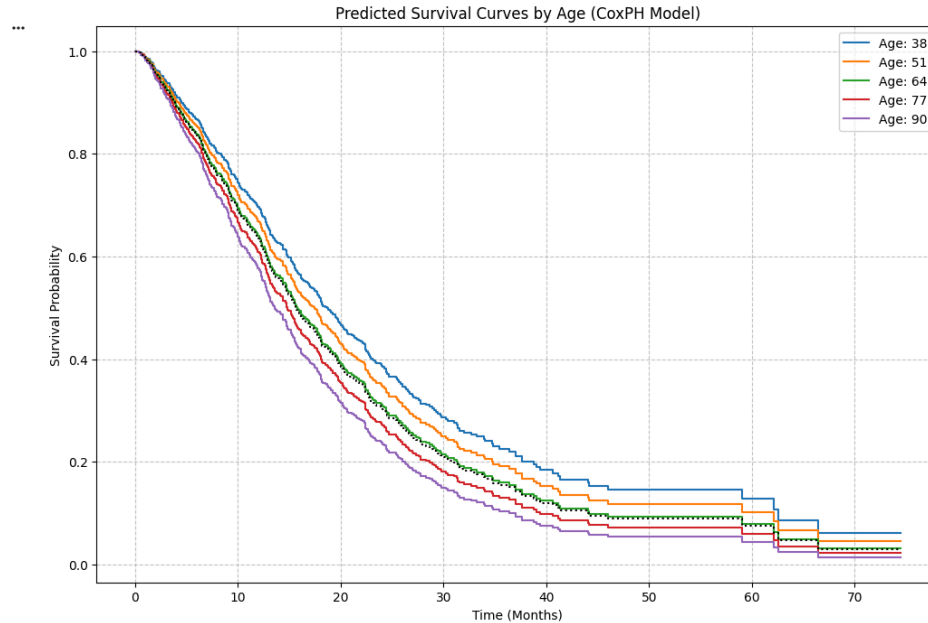
## Continuous Survival Analysis and Hazard Metrics

Initially, a Kaplan-Meier survival analysis by age quartiles was created and demonstrated massive overlap between the curves (Figure 4). This did not succeed in establishing distinct risk delineations between age groups.



**Figure 4. Kaplan–Meier Survival Curves Stratified by Age Quartiles.** Categorical survival analysis partitioning patients into four age-based quartiles (Q1–Q4). The extensive overlap in confidence intervals demonstrates the inability of discrete age categorization to show distinct risk profiles.

To resolve this, a semi-parametric proportional hazards model was deployed with a Breslow baseline estimation across the 670 records. Unlike the previous model that had static quartile curves, this one presented step-by-step risk gradients across the timeline (Figure 5). This revealed the true shifts in survival probability from age 38 through 90.



**Figure 5. Predicted Continuous Survival Curves from the Multivariable Cox Proportional Hazards Model.** Stepwise continuous survival curves generated at representative age intervals (ages 38, 51, 64, 77, and 90) using the Cox proportional hazards model, demonstrating refined risk stratification across the continuous age continuum.

When evaluated as an unmanipulated, continuous covariate alongside clinical factors, Age at Diagnosis displayed a stable continuous hazard profile ( $\{\exp(\text{coef})\} = 1.01$ , 95% CI: [1.00, 1.02],  $p = 0.09$ ). However, multivariable analysis identified structural and genomic irregularities as the primary drivers of mortality risk within the group (Table 1)



|                                  | coef  | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z     | p      | -log2(p) |
|----------------------------------|-------|-----------|----------|----------------|----------------|---------------------|---------------------|--------|-------|--------|----------|
| Age at Diagnosis                 | 0.01  | 1.01      | 0.00     | -0.00          | 0.02           | 1.00                | 1.02                | 0.00   | 1.70  | 0.09   | 3.48     |
| Mutation Count                   | -0.01 | 0.99      | 0.07     | -0.15          | 0.13           | 0.86                | 1.14                | 0.00   | -0.16 | 0.87   | 0.20     |
| Fraction Genome Altered          | 0.89  | 2.43      | 0.33     | 0.23           | 1.54           | 1.26                | 4.66                | 0.00   | 2.66  | 0.01   | 7.00     |
| TMB (nonsynonymous)              | 0.02  | 1.02      | 0.08     | -0.14          | 0.17           | 0.87                | 1.19                | 0.00   | 0.20  | 0.84   | 0.25     |
| Sex_Male                         | 0.18  | 1.20      | 0.09     | 0.00           | 0.37           | 1.00                | 1.44                | 0.00   | 2.00  | 0.05   | 4.45     |
| Ancestry_African (AFR)           | 0.25  | 1.28      | 0.25     | -0.25          | 0.74           | 0.78                | 2.10                | 0.00   | 0.97  | 0.33   | 1.60     |
| Ancestry_East Asian (EAS)        | 0.09  | 1.10      | 0.25     | -0.39          | 0.58           | 0.68                | 1.78                | 0.00   | 0.37  | 0.71   | 0.49     |
| Ancestry_European (EUR)          | -0.10 | 0.91      | 0.16     | -0.40          | 0.21           | 0.67                | 1.23                | 0.00   | -0.63 | 0.53   | 0.92     |
| Ancestry_South Asian (SAS)       | 0.20  | 1.22      | 0.57     | -0.92          | 1.31           | 0.40                | 3.71                | 0.00   | 0.35  | 0.73   | 0.46     |
| KRAS Somatic Variant_G12D        | 0.18  | 1.20      | 0.37     | -0.54          | 0.91           | 0.58                | 2.48                | 0.00   | 0.50  | 0.62   | 0.70     |
| KRAS Somatic Variant_G12R        | -0.21 | 0.81      | 0.38     | -0.95          | 0.54           | 0.39                | 1.71                | 0.00   | -0.54 | 0.59   | 0.77     |
| KRAS Somatic Variant_G12V        | 0.08  | 1.09      | 0.37     | -0.64          | 0.81           | 0.53                | 2.25                | 0.00   | 0.22  | 0.82   | 0.28     |
| KRAS Somatic Variant_Other       | -0.05 | 0.95      | 0.43     | -0.90          | 0.79           | 0.41                | 2.21                | 0.00   | -0.12 | 0.90   | 0.15     |
| KRAS Somatic Variant_Q61H        | -0.19 | 0.82      | 0.43     | -1.04          | 0.65           | 0.35                | 1.92                | 0.00   | -0.45 | 0.66   | 0.61     |
| KRAS Somatic Variant_Q61R        | 0.51  | 1.67      | 0.48     | -0.43          | 1.46           | 0.65                | 4.30                | 0.00   | 1.06  | 0.29   | 1.80     |
| Copy Number State_Balanced       | -0.66 | 0.52      | 0.21     | -1.07          | -0.26          | 0.34                | 0.77                | 0.00   | -3.21 | <0.005 | 9.54     |
| Copy Number State_CNLOH          | 0.12  | 1.13      | 0.26     | -0.40          | 0.64           | 0.67                | 1.90                | 0.00   | 0.46  | 0.64   | 0.63     |
| Copy Number State_Gain           | -0.28 | 0.76      | 0.22     | -0.70          | 0.15           | 0.50                | 1.16                | 0.00   | -1.28 | 0.20   | 2.32     |
| Copy Number State_LOH            | -0.81 | 0.45      | 0.24     | -1.28          | -0.33          | 0.28                | 0.72                | 0.00   | -3.34 | <0.005 | 10.20    |
| Copy Number State_Loss after WGD | -0.49 | 0.61      | 0.27     | -1.02          | 0.04           | 0.36                | 1.04                | 0.00   | -1.83 | 0.07   | 3.89     |
| MSI Type_Indeterminate           | -0.08 | 0.93      | 0.47     | -1.00          | 0.85           | 0.37                | 2.33                | 0.00   | -0.16 | 0.87   | 0.20     |
| MSI Type_Unstable                | -1.66 | 0.19      | 1.34     | -4.29          | 0.96           | 0.01                | 2.61                | 0.00   | -1.24 | 0.21   | 2.22     |
| MSI Type_Stable                  | -0.07 | 0.93      | 0.16     | -0.40          | 0.25           | 0.67                | 1.28                | 0.00   | -0.45 | 0.65   | 0.62     |
| TMB-H_High(>=10)                 | -0.33 | 0.72      | 0.65     | -1.61          | 0.95           | 0.20                | 2.58                | 0.00   | -0.51 | 0.61   | 0.71     |

**Table 1. Multivariable Cox Proportional Hazards Model for Overall Survival.**

Summary of parameter estimates, hazard ratios, standard errors, 95% confidence intervals, and significance levels evaluating continuous age at diagnosis alongside somatic mutation profiles, copy-number states, and microsatellite instability (MSI) status (N = 670). Continuous age displayed a modest hazard profile ( $p = 0.09$ ), while structural genomic features such as Fraction Genome Altered ( $p = 0.01$ ) and Loss of Heterozygosity ( $p = 0.005$ ) served as primary independent drivers of mortality risk.

The model achieved a concordance index of 0.61 with a partial AIC of 5,772,37. The log-likelihood ratio test (72.02 on 24 degrees of freedom,  $-\log_2(p) = 19.82$ ) confirmed that integrating molecular alterations along with continuous age variables significantly enhances risk prediction accuracy compared to the baseline clinical criteria alone.

## Discussion

This study looked to identify how somatic mutation frequencies and patient survival outcomes change across a continuous age spectrum in patients with Pancreatic Ductal Adenocarcinoma (PDAC). Additionally, it looked to solve whether continuous modeling would uncover clinical and molecular patterns initially not seen because of traditional categorical age binning.

Using clinical and genomic profiles from 670 PDAC patient records, age at diagnosis was analyzed as an unmanipulated continuous independent variable. Non-linear proportion mapping tracked KRAS driver variant shifts across continuous decade points. Survival dynamics were evaluated using unbinned Cox Proportional Hazards alongside categorical Kaplan-Meier curves. Multivariate linear regression was used to evaluate genomic feature predictions.

The results demonstrate that treating patient age as a continuous variable reveals non-linear shift and survival dynamics in PDAC that are obscured by traditional categories. The observed divergence between KRAS G12D and G12V across the continuous age decades suggests age-dependent selection mechanisms within the pancreatic environment.

Furthermore, comparing the binned Kaplan-Meier curves against the continuous Cox models confirms that age categorization can mask underlying risk factors. While age itself showed a modest hazard profile when controlling for genomic features, structural driver alterations like Fraction Genome Altered and Copy Number LOH served as the most relevant predictors of overall survival. The low performance of linear baseline models shows the non-linear nature of pancreatic cancer genomic features.

This study is mostly limited by sample density at extreme age ranges and feature collinearity in the linear modeling. Future work should utilize non-linear regression models and larger populations that will further refine these continuous lifespan models. Continuous age modeling should also be extended to other gastrointestinal and solid tumors to determine if G12D and G12V divergence is tissue-specific or a feature of somatic aging.

Overall, adopting continuous modeling approaches is essential in biology when studying cancer genomics. Evaluating somatic mutations and survival trajectories along a continuous timeline provides a much clearer understanding of how aging influences tumor evolution and patient risk profiles.

## **Conclusion**

This study demonstrates the necessity of continuous analytical frameworks in cancer genomics. By avoiding the usual age categorization, a clear non-linear divergence in KRAS somatic driver variants across an adult lifespan was revealed. This finding was characterized by a steady decline in G12D dominance alongside a continuous increase in the prevalence of G12V. Furthermore, continuous survival modeling showed precise risk trajectories that resolved the heavy overlaps observed in the traditional binned quartile curves. Structural genomic alterations, like Fraction Genome Altered and Loss of Heterozygosity, were the main mortality drivers when accounting for continuous age. The incorporation of continuous lifespan models in genomic research preserves biological signals, improves risk stratification, and offers a needed baseline for age-based care in oncology.

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